

API and REST API

What is an API?

An API (Application Programming Interface) is a set of rules that allows different software applications to communicate with each other. It acts as a bridge between systems, enabling them to exchange data and services without needing to know how the other system is internally built.

Examples:

- A weather app uses an API to get weather data from a weather server.
- Mobile applications use APIs to connect to databases and online services.
- Payment systems like PayPal or Stripe provide APIs for online transactions.

What is a REST API?

A REST API (Representational State Transfer API) is a type of API that follows REST architectural principles. REST APIs use standard HTTP methods to perform operations on resources such as users, products, or orders.

REST APIs are:

- Simple and easy to use
- Stateless
- Usually exchange data using JSON
- Widely used in web and mobile development

Common HTTP Methods in REST API

1. GET

Used to retrieve data from the server.

Example: GET /users

2. POST

Used to send new data to the server and create a new resource.

Example: POST /users

3. PUT

Used to update existing data completely.

Example: PUT /users/1

4. DELETE

Used to remove data from the server.

Example: DELETE /users/1

5. PATCH

Used to update part of the data instead of replacing everything.

Example: PATCH /users/1

Advantages of REST APIs

- Easy to understand and implement

- Platform independent
- Fast communication using HTTP
- Supports multiple data formats
- Widely supported in modern applications

Conclusion

APIs allow software systems to communicate efficiently. REST APIs are popular because they are lightweight, flexible, and easy to use. HTTP methods such as GET, POST, PUT, DELETE, and PATCH help developers manage data in web applications.